Parameter sensitivity of federation benefit I

Studies 1–7 fix specific parameter configurations (sensor acuity, colony size) to isolate mechanistic claims. A natural question is whether the federation benefit is **robust to those choices** or is an artifact of a narrow operating point. Study 8 addresses this with a systematic 2-D sensitivity sweep over sensor acuity and colony size.

We sweep sensor acuity $\kappa \in \{0.40, 0.55, 0.70, 0.85, 0.95\}$ and colony size $n \in \{2, 4, 6, 8, 10\}$, evaluating two systems:

Parameter sensitivity of federation benefit I

- **Belief sharing** (Study 1 architecture) — accuracy gap = communicating minus isolated mean accuracy;
- **Hierarchical POMDP** (Study 6 architecture) — accuracy gap = hierarchical minus flat location accuracy.

Parameter sensitivity of federation benefit I

Each cell averages $n_{\text{trials}} = 20$ independent trials to reduce Monte-Carlo noise at the cell level.

The resulting heatmaps (fig. 1) show the accuracy gap for both systems as a function of acuity and colony size. Green cells indicate that federation benefits the colony; red cells indicate that the chosen configuration yields no benefit or a slight deficit. The symmetric RdYlGn colormap is centered on zero so the sign of the benefit is immediately legible.

Parameter sensitivity of federation benefit I

Across the grid the following patterns hold:

Parameter sensitivity of federation benefit I

1. **The belief-sharing benefit lives at low-to-moderate acuity.** The accuracy gap peaks in the second-lowest acuity row — where individual observations carry some signal but no single sentinel resolves the location alone, so pooled evidence pays most — and remains uniformly positive across the two lowest-acuity rows for colonies of at least four agents. Near ceiling acuity the gap shrinks toward zero: single agents already solve the task, so federation has nothing left to add.
2. **Colony size acts through a floor, not a smooth slope.** The two-agent column shows an exactly zero belief-sharing gap at every acuity: under self-exclusion ("agents do not hear themselves"), each member of a two-agent colony hears exactly one incoming belief, so the heard consensus adds no pooled evidence. Colonies of four or more realize the low-acuity benefit.
3. **The hierarchical gap is near zero across most of the grid.** The hierarchical-minus-flat location-accuracy gap is approximately zero over most cells, with a few strongly negative low-acuity cells and small positive cells confined to the two-agent column — consistent with the per-study finding that the hierarchical architecture matches rather than beats the flat baseline on location accuracy.

Parameter sensitivity of federation benefit I

The full parameter grid and protocol details are in the supplement (sec. 13.14). A native-unit cross-study overview of the headline metrics across all 9 studies is shown in fig. 2.

Parameter sensitivity of federation benefit I

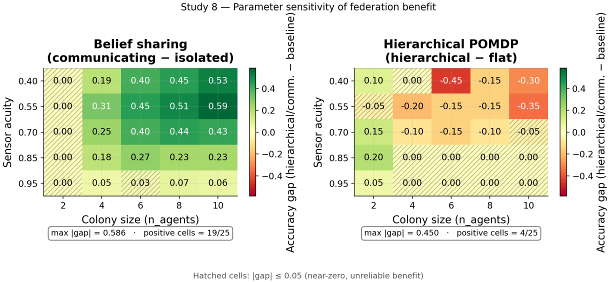

Figure 1: Source relation: original project sensitivity diagnostic; estimand: per-cell accuracy gaps (fractions) as functions of acuity and colony size; uncertainty: deterministic per-cell means over the declared trials, with no resampling interval. Two-panel heatmap (1×2) of the Study 8 parameter sensitivity sweep. Left panel: y-axis indexes sensor acuity (0.40–0.95, 5 levels); x-axis indexes colony size (2–10 agents, 5 levels); color encodes the belief-sharing accuracy gap (communicating minus isolated mean accuracy). Right panel: identical axes; color encodes the hierarchical POMDP location accuracy gap (hierarchical minus flat). Color scale: RdYlGn symmetric around zero — green denotes federation benefit, red denotes deficit. Diagonal hatching marks cells with $|\mathrm{gap}| \leq 0.05$ (unreliable, near-zero benefit). Cell values are deterministic per-cell means over 20 trials; no resampling error band is shown — the sweep protocol is detailed in the sensitivity supplement.

Parameter sensitivity of federation benefit II

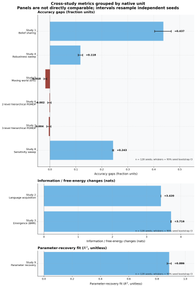

Figure 2: Source relation: original project cross-study summary; estimand: grouped study-level means in native units (accuracy fractions, nats, or R^2), never a cross-unit ranking; uncertainty: seed-level bootstrap confidence intervals. Horizontal native-unit facet chart summarizing the key federation benefit metric for each of the 9 studies (Studies 1–9). x-axis indexes benefit value in metric-specific units (accuracy gain for Studies 1, 4, 5, 6, 7, 8; KL reduction for Study 2; ΔF for Study 3; R^2 for Study 9). y-axis lists the 9 studies (one row each), ordered from Study 1 at the top to Study 9 at the bottom within each native-unit facet. Each mark shows the mean over 128 independent seeds with intervals spanning the 95 % bootstrap confidence interval. There is no cross-unit ranking: zero is a within-unit reference only, and the facet labels carry the units. Consistent with the per-study results, Studies 5 (moving world, EFE) and 6 (2-level hierarchical) sit at approximately zero within their respective units.